

SIMATS ENGINEERING

NAME :Swetha.B

Reg no.: 192412215

ITA0606-

Machine Learning

Experiment -12

Implement Iris Flower Classification using KNN

Code:

```
# Iris Flower Classification using K-Nearest Neighbours (KNN)

import pandas as pd

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report

# Load Iris dataset
iris = load_iris()

X = iris.data
y = iris.target

# Split dataset into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature scaling
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Create KNN model
k = 5
knn = KNeighborsClassifier(n_neighbors=k)

# Train the model
knn.fit(X_train, y_train)
```

```
# Make predictions
y_pred = knn.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)

print("Iris Flower Classification using KNN")
print("-----")
print("K value:", k)
print("Accuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Classification Report
print("\nClassification Report:")
print(classification_report(
    y_test,
    y_pred,
    target_names=iris.target_names
))

# Actual vs Predicted
result = pd.DataFrame({
    "Actual": [iris.target_names[i] for i in y_test],
    "Predicted": [iris.target_names[i] for i in y_pred]
})

print("\nActual vs Predicted:")
print(result)
```

OUTPUT:

*** Iris Flower Classification using KNN

K value: 5
Accuracy: 100.00%

Confusion Matrix:
[[10 0 0]
[0 9 0]
[0 0 11]]

Classification Report:

	precision	recall	f1-score	support
setosa	1.00	1.00	1.00	10
versicolor	1.00	1.00	1.00	9
virginica	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

Actual vs Predicted:

	Actual	Predicted
0	versicolor	versicolor
1	setosa	setosa
2	virginica	virginica
3	versicolor	versicolor
4	versicolor	versicolor
5	setosa	setosa
6	versicolor	versicolor
7	virginica	virginica
8	versicolor	versicolor
9	versicolor	versicolor
10	virginica	virginica
11	setosa	setosa
12	setosa	setosa
13	setosa	setosa
14	setosa	setosa
15	versicolor	versicolor
16	virginica	virginica
17	versicolor	versicolor
18	versicolor	versicolor
19	virginica	virginica
20	setosa	setosa
21	virginica	virginica
22	setosa	setosa
23	virginica	virginica
24	virginica	virginica
25	virginica	virginica
26	virginica	virginica
27	virginica	virginica
28	setosa	setosa
29	setosa	setosa